CSE 6179 – Digital Forensics

Class Assignment # 1 RAM Imaging & Information Extraction Spring 2026

For this class assignment, complete the following activities in your PC:

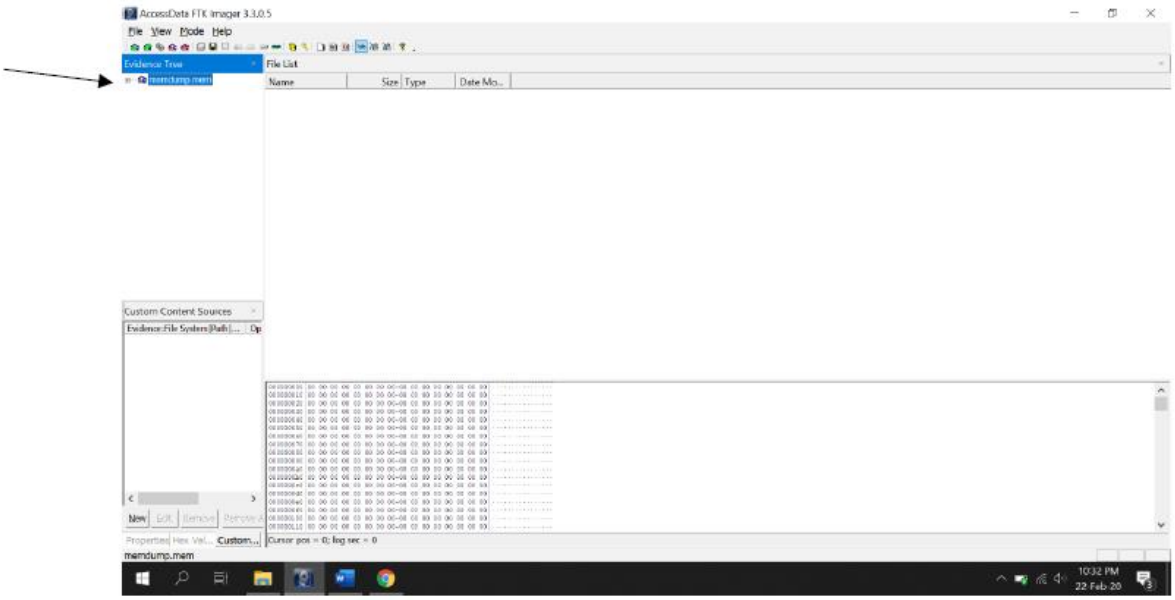
1. **What were you Googling? (**In this task, we will prove that someone was **actively** looking for some ransomware attacks and viewed one of the searching results.)
   a. Go to Google and search for "**ransomware attacks**"
   b. Then **click the first link** on the search result (Wikipedia)
2. **Detecting Facebook Activities – Searching within FB and Viewing Videos** (We have captured Google search in earlier activity but what about the search within the Facebook, clicking and viewing on some videos. Let's find out if we can grab the info about what a suspect was looking for in facebook)
   a. Go to **facebook.com** and search for "**United International University**"
   b. **Click** and **watch** any one of the **videos** uploaded in the page.
   c. For investigation, one of the activities that could be detected is **viewing other's profiles** by the suspect. **Click and go** to any profile page for the experimental data.
3. **Capturing Gmails Communications!** (Last but not least, **Bulk_Extractor** is capable not only of capturing **social media events**, but also any **sensitive information** like credit cards. In this task a private Gmail emails, **send a test email** from your gmail account to any other account that can be grabbed and analyzed.
4. You can **maintain confidentiality** not to expose any sensitive data during the work.

Q.1 Capture RAM of your PC by using **FTK Imager** Software. Specify the steps with screenshots.

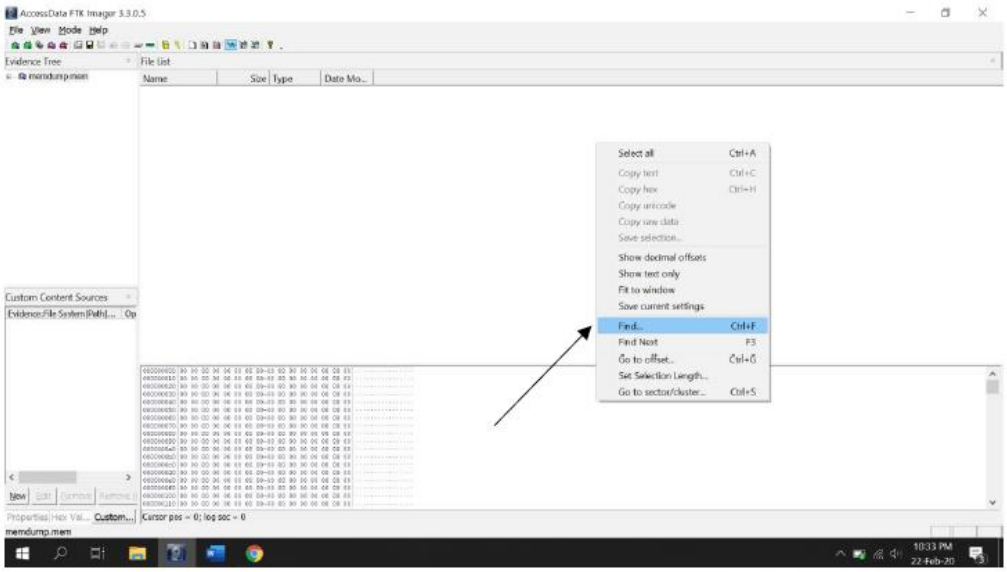
Q.2 Analyze the created **RAM's image** by using **FTK Imager** and find out **web related information**.

Instructions

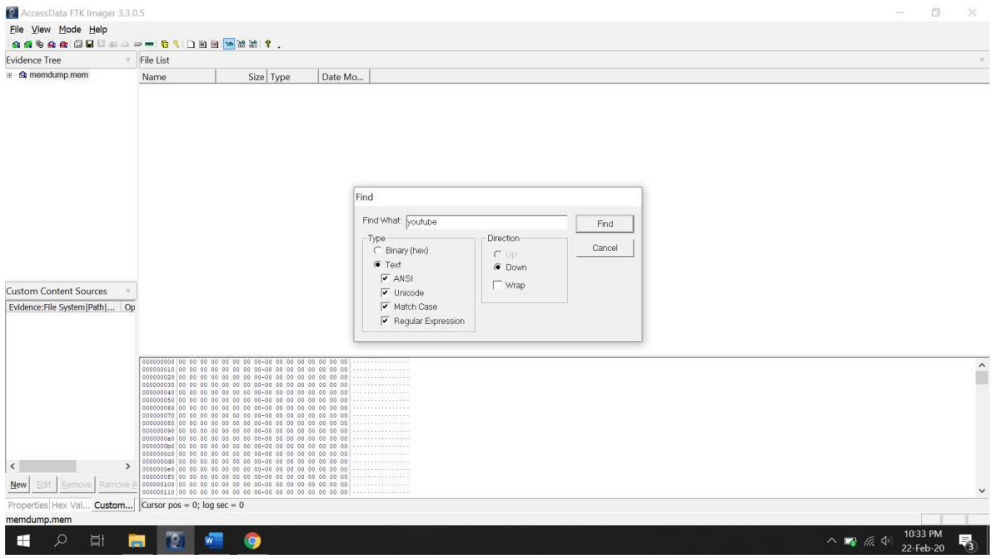
1. Open FTK Imager Software on the PC.
2. From "File" option select "Add Evidence Item".
3. In the "Select Source" box, click "Image File" followed by "Next".
4. Search for the RAM Image file created earlier (**Q.1**) in the preferred destination and then click "Finish".
5. From the "evidence tree" section, select the image file.

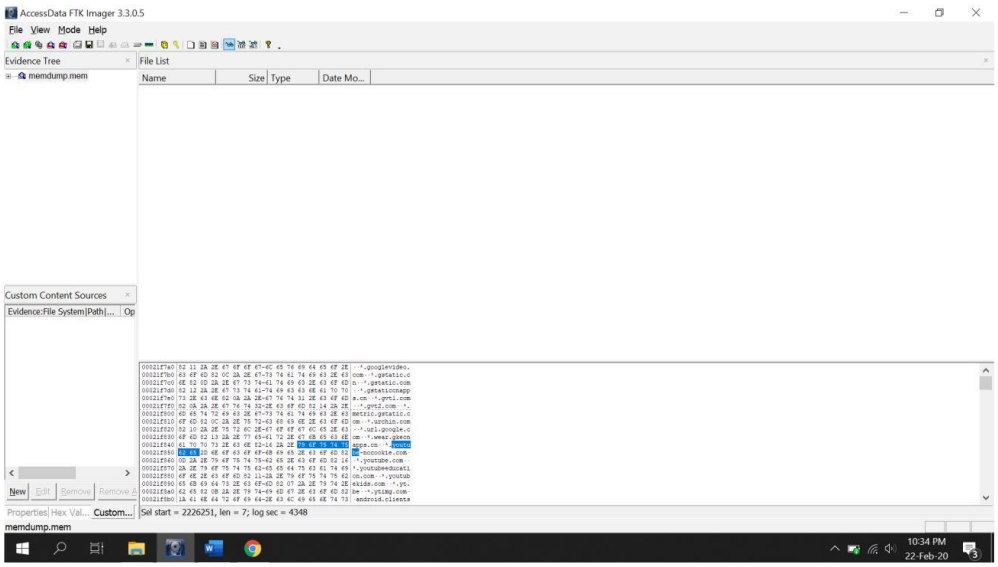


6. From the section with hexadecimal entries, right click to initiate search option.

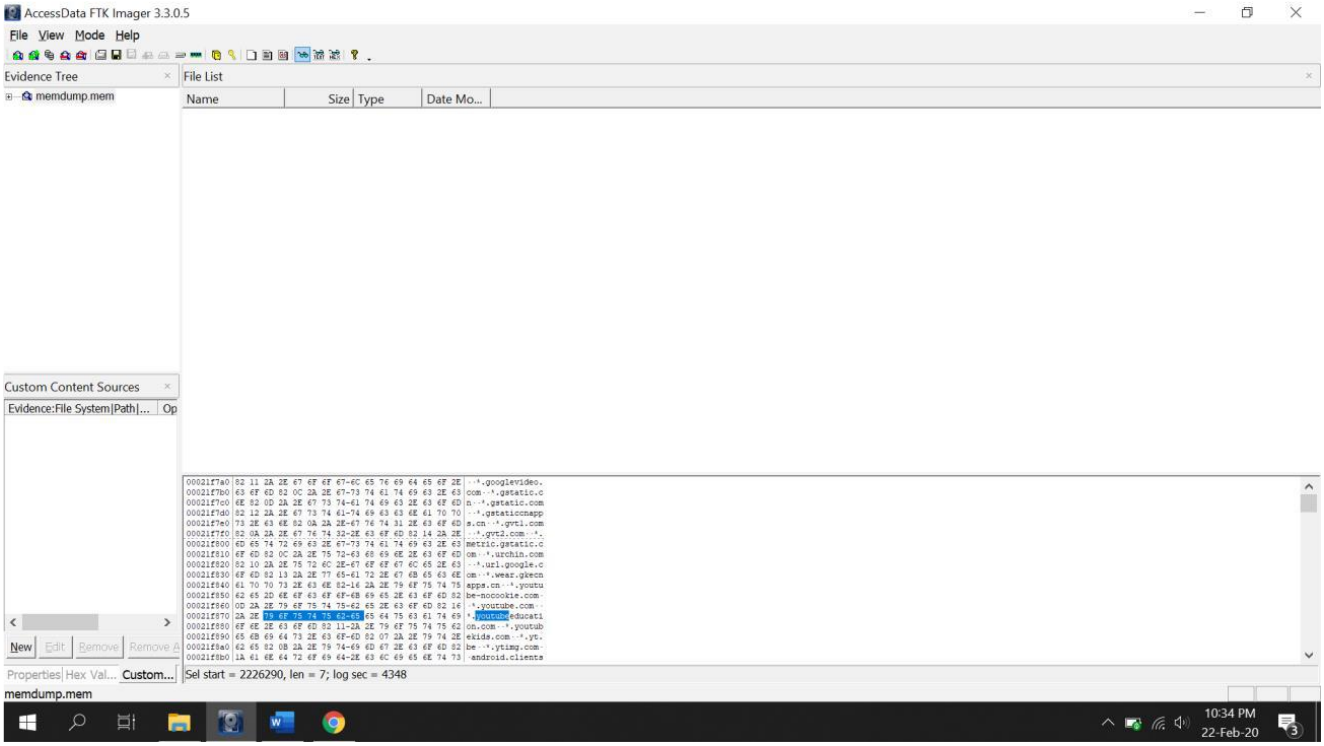


7. Enter website name to get detailed information.





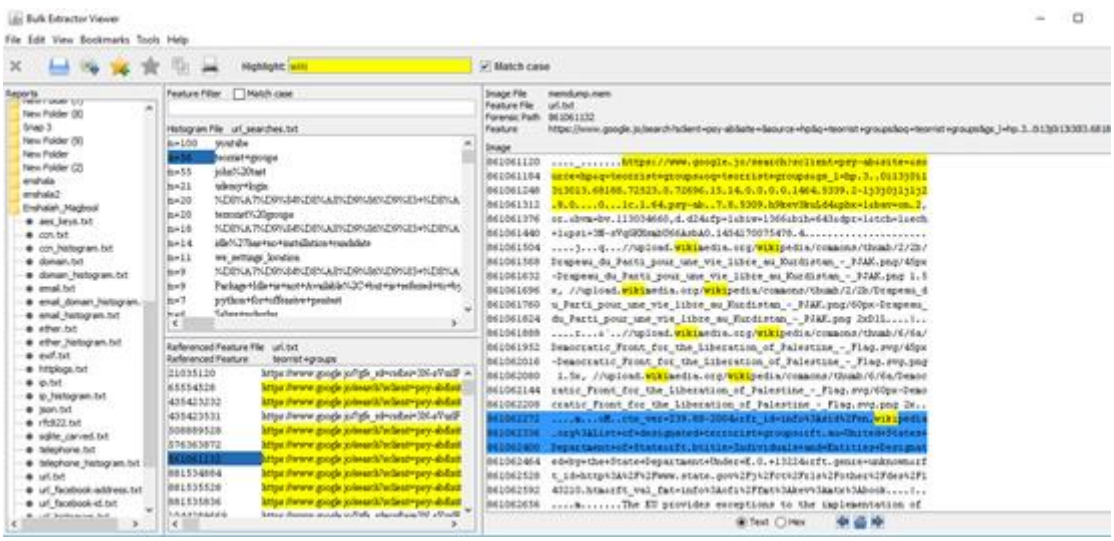
8. On pressing F3 we get more information about that particular website.



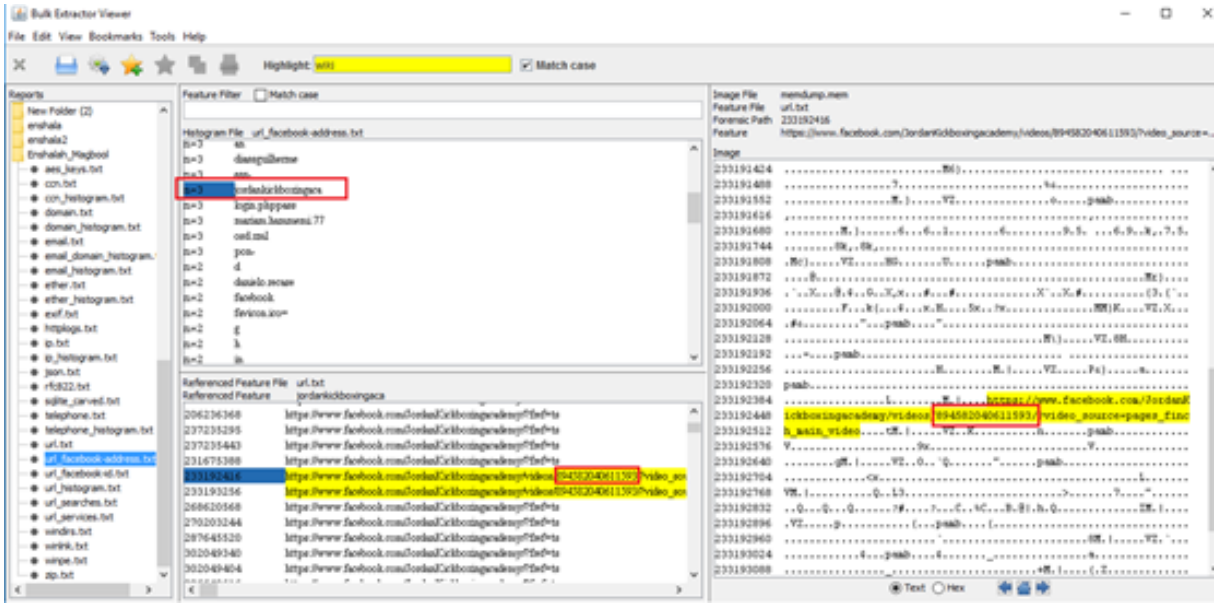
9. Find a **list of 5 web sites** browsed from the PC.

Q.3 Analyze the created **RAM's image** by using **Bulk Extractor** and find out **following information** collected by different scanners.

1. (What were you Googling?): After importing the memory dump from the suspect machine, Bulk_Extractor will generate its report and a list of correlated features files. One of them called "**URL_Searches.txt**", and it's to responsible for tracking your searching activities, and it's obvious that our suspect has searched for "terrorist" and when we highlight for "wiki" keyword, we see that his visited the Wikipedia for a "**List_of_designated_terrorist_groups**". (See following figure)

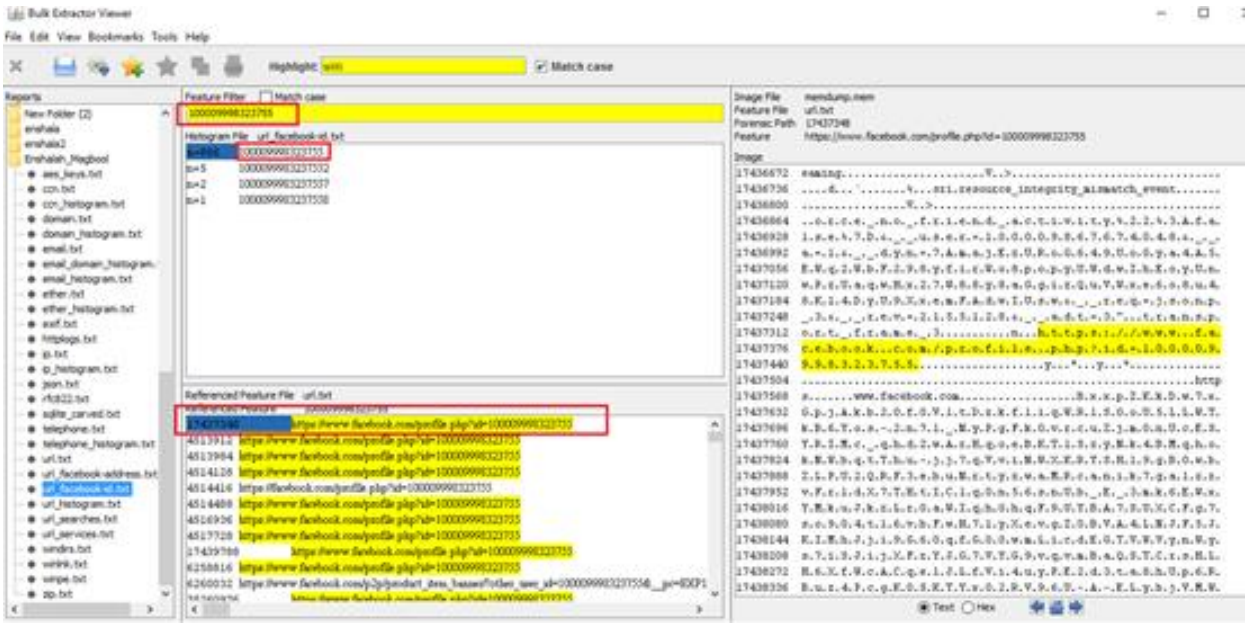
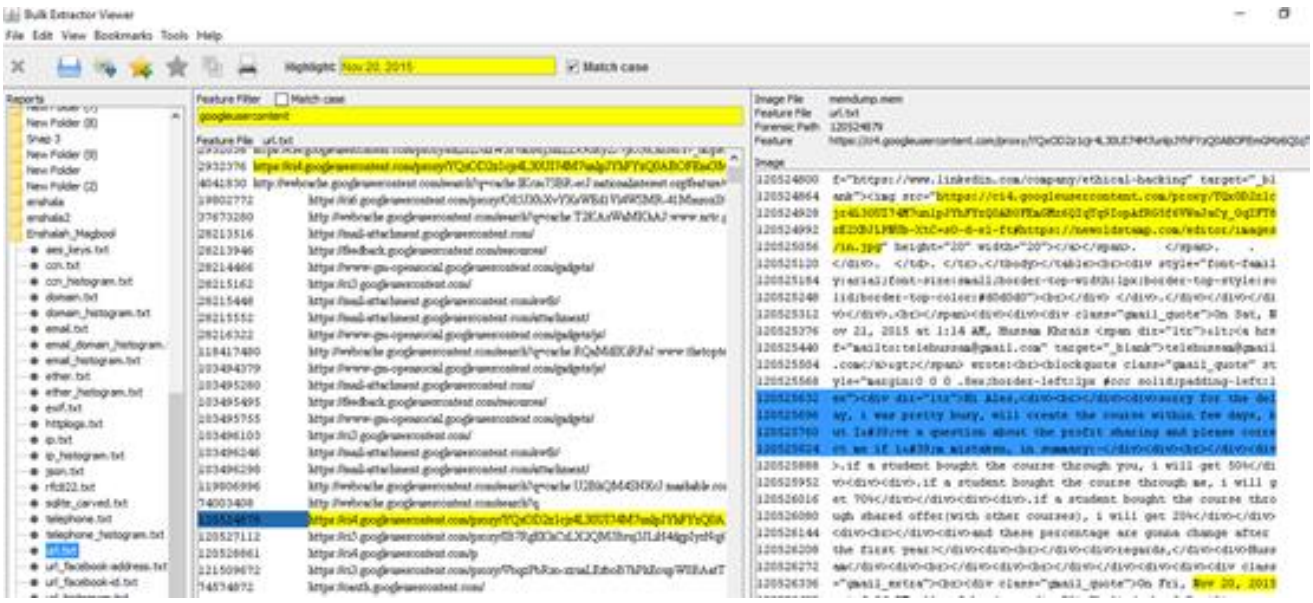


2. (Detecting Facebook Activities – Searching within FB and Viewing Videos): Thankfully, Bulk_extractor makes our life much easier by addressing such activities in clear format inside "**URL_facebook-addresses**." (See following figure)



If we pay some attention to memory dump, we can reassemble the raw data into a direct link pointing to video that the suspect has been watching.

And that action was logged, and we can easily pick it up with quick searches in "**URL_facebook-id.txt**".



3. (Capturing Gmails Communications!) Now within quick search in "**URL.txt**" and filter on "**googleusercontent**," we were able to recover the **message along with the sent date** as we can see down below.